

The model harness is used for testing, and SIL is now reused as a test execution framework for the processor board. With PIL, tests are executed while having co-simulation of the embedded algorithm compiled and deployed on the target processor board and the existing executable specification.

Besides tests for functional equivalence, PIL also lets us step through assembly-level instructions in the debugger and analyse codes that were compiled, linked and deployed, just as these will run on the real system.

With PIL one can also review the order of execution of code functions as well as verify calls to OS functions, or other libraries required for execution. On the target, it is also possible to monitor the memory footprint during execution of verification scenarios. PIL provides an opportunity to compare software algorithm behaviour on different target processors or boards, for example, from different vendors.

Hardware-in-the-loop

Verification and testing methods described above rely on non-real-time based simulations and therefore cannot be used to verify real-time aspects of the design. The overhead of simulation and communication with the target processor do not permit real-time testing of the software algorithm. Hardware-in-the-loop (HIL) allows you to test the software algorithm with real-time behaviour.

With HIL, you execute models in real-time on a PC based target computer using a library of I/O card drivers, real-time kernel and host-target interface for real-time monitoring, parameter tuning and data logging. Such configuration reduces the risk of testing on the actual, and often expensive, device (when available). This type of verification needs sophisticated signal conditioning and power electronics to properly stimulate inputs and receive outputs of the target hardware. HIL is typically done in the lab as a final test

before system integration and field testing.

Conclusion

MBD with graphical model simulation and code generation provides powerful capabilities for embedded software development. It allows research and development teams to explore design and implementation alternatives. With model testing, the design is validated in simulation using input test vectors and expected results. Once the design is validated in simulation, it is converted to C/C++ codes via automatic code generation.

SIL testing validates that the generated codes have the same behaviour as model simulation. PIL testing is used to verify bit-accurate execution on the target processor to ensure that simulation and executable codes are within specified tolerances.

Finally, HIL permits testing of the software algorithm with real-time behaviour. ●



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